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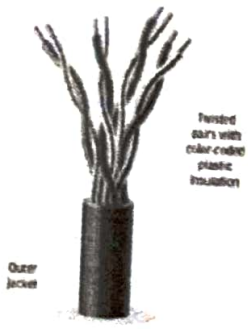
2. Study of different types of network cables

- **AIM:** To study and understand different types of network cable. To make your own Ethernet Cross-over Cable / straight cable.

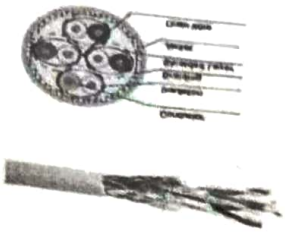
a) **Understand different types of network cable:**

1. Unshielded Twisted pair (UTP) Cable
2. Shielded Twisted pair (STP) Cable
3. Coaxial cable
4. Fibre optic cable

cable type	category	Maximum Data transmission	Advantages / Disadvantages	Application / Use
UTP	Category 3	10 bps	**Advantages:** • cheaper in cost • Easy to install as they a smaller overall diameter **Disadvantages:** • More prone to (EMI) Electro mag-netic interface and noise	10Base-T Ethernet
	Category 5	Upto 100 MBPS		Fast Ethernet Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet Gigabit Ethernet
STP	Category 6. 6a	10 Gbps	**Advantages:** • Shielded • Faster than UTP • Less susceptible to noise and interface	Gigabit Ethernet 10G Ethernet (55 m) widely used in data centers.


Twisted pairs with color-coded plastic insulation
Outer jacket

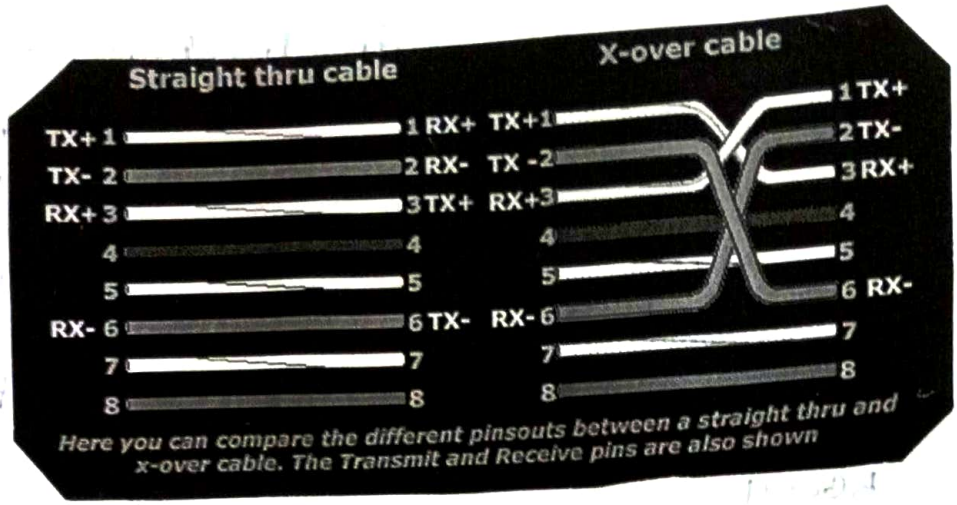

Shielded twisted pair



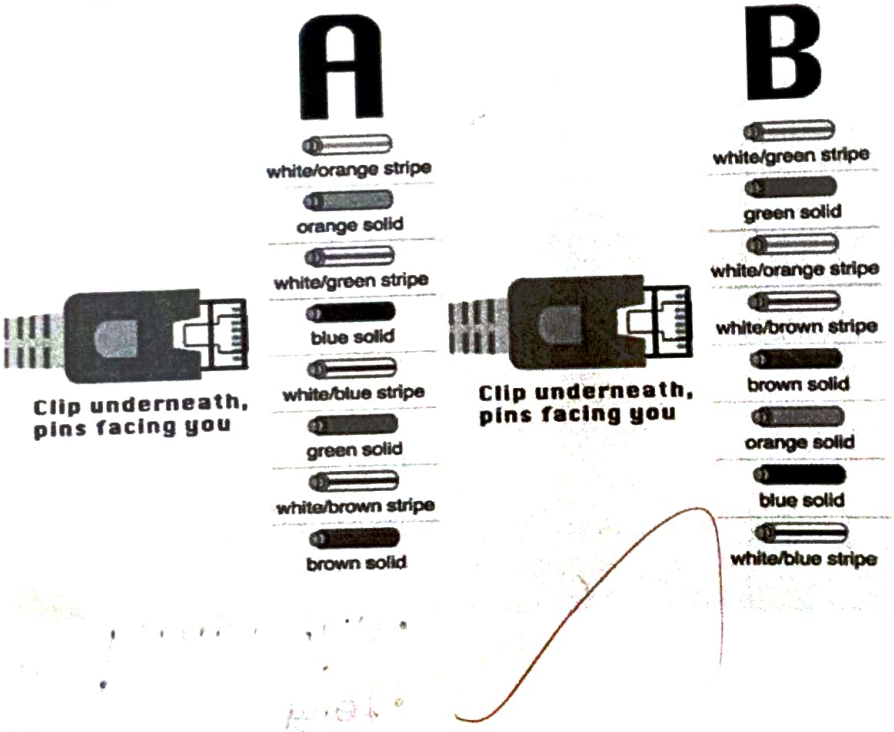

Braided Shield
Foil Shield
Center Conductor
Outer Jacket
Dielectric



cable type	category	Maximum Data transmission	Advantages / Disadvantages	Application / Use
SSTP	Category 7	10Gbps	**disadvantages:** • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (100m)
Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	**Advantages:** • High bandwidth • Immune to interference • low loss bandwidth • versatile **Disadvantages:** • limited distance • Cost • size is bulky	Speed of signal is 500m Television network High Speed internet connections
fibre optics cable	Single mode Multi mode	100 Gbps	**Advantages:** • High speed • High bandwidth • High security • long distance	Maximum distance of fibre optics cable is around 100 meters



Straight through network cable: both sides should be A
Crossover cable: One side A, one side B



Cable type	category	Maximum Data transmission	Advantages/ Disadvantages	Appl Use
			Disadvantages: • Expensive • Requires skilled installers	

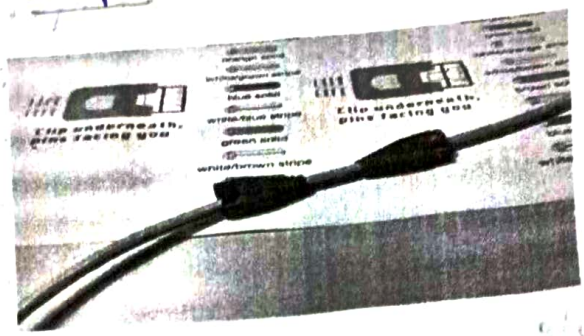
Make your own Ethernet Cross-Over Cable / Straight Cable

Tools and parts needed:

- Ethernet Cable: CAT5e is certified for gigabit support, but CAT 5 cabling works as well, just over shorter distance
- A crimping tool: This is an all-in-one networking tool shaped to push
- Two RJ45 plugs
- Optional two plug shields

Year / Branch / Section :

Step 1 :



Step 2 :



Step 3 :



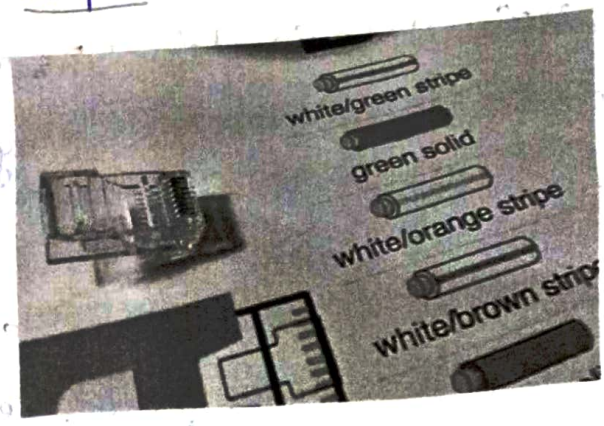
The steps include :

- Step 1 : To start construction of the device . Begin by threading shields onto the cable
- Step 2 : Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.
- Step 3 : After , you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom One end should be in arrangement A and the other in B.
- Step 4 : Once the order is correct, bunch them together in a line , and if there are any that stick out farther than others, snip them back to create an even level The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you , as shown
- Step 5 : Next, push the cable right in . The notch at the end of the plug needs to be just over the cable shielding and if it isn't , that means that you stripped off too much shielding. Simply snip the cables back a little more.

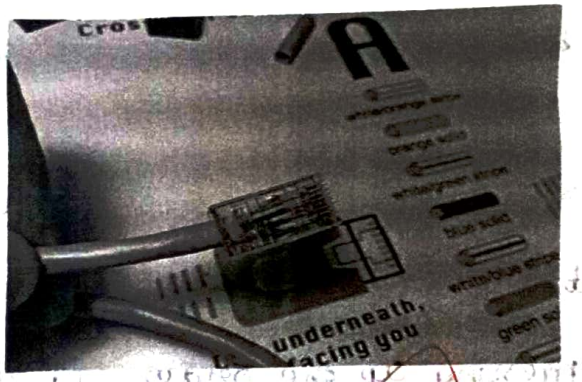
Register No. :

Name / Branch / Section :

Step 4:



Step 5:



- Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down It should be shaped correctly, but pushing too hard can crack the fragile plastic plug
- Step 7: Lastly, repeat for the other end using diagram B C to make a crossover cables) / using diagram A (to make a straight through cable) To test it, plug it in and attempt to connect two devices directly.

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- RESULT: Thus, different types of network cables were studied successfully, and straight through and cross over Ethernet cables were prepared and tested successfully.

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